

Prashant Katiyar

*Head of AI & Applied Technology
GenAI, Computer Vision & NLP Expert /
Multi-Agent Systems Specialist*

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Professional Summary

Visionary AI leader with 7+ years of hands-on experience designing, building, and leading end-to-end AI systems from research to production. Currently Heading AI at SafetyWhat AI, where I own the full AI stack — multimodal LLMs, LangGraph/AutoGen multi-agent systems, RAG pipelines, computer vision models, and scalable inference on Kubernetes/Triton. Proven track record of delivering production-grade, zero-downtime agentic workflows and state-of-the-art computer vision systems that reduce compute costs by 3x+ while maintaining or improving accuracy. Deep expertise across foundational models, transformers, fine-tuning, RAG, agentic architectures, MLOps, cloud-native deployment, and leading cross-functional/global teams.

Core Competencies & Technical Skills

Python (expert), C++, C#, PyTorch, TensorFlow, JAX, Hugging Face, LangChain, LangGraph, AutoGen, LlamaIndex, Spacy, NLTK, BERT/GPT variants, OpenRouter, Modal, Haystack, NVIDIA Triton, TensorRT, LLMs (fine-tuning, LoRA/QLoRA, RAG, agentic flows), Transformers, Foundational Models, Computer Vision, NLP, Multimodal AI, OCR (Tesseract, EasyOCR, PaddleOCR), PDF processing (PyMuPDF, LayoutParser), Vector DBs, Unstructured pipelines, AWS (Bedrock, SageMaker, ECS, Lambda, S3), GCP Vertex AI, Kubernetes, Docker, MLOps: MLflow, CI/CD, model monitoring, Django, FastAPI, PostgreSQL, Redis, Elasticsearch, CUDA, OpenGL, DeepSpeed, FSDP

Professional Experience

April 2023 – **Head of AI & Applied Technology, SafetyWhat AI, India** (global remote teams)

- Present - Lead entire AI organization's strategy & execution; own roadmap, architecture, and delivery of production GenAI + CV systems serving 100K+ daily users with zero-downtime requirements
- Designed and deployed multi-agent systems using LangGraph + Shimmy + Llama-3 with RAG over PDFs, sensor data → 92%+ automation in diagnosis reporting
- Built end-to-end RAG pipelines for unstructured safety reports and video transcripts (EasyOCR + LayoutParser + LLM) → structured JSON
- Engineered scalable inference on Kubernetes + NVIDIA Triton with auto-scaling, canary deployments, model versioning, and active monitoring
- Fine-tuned Mistral-7B, Phi-3, Llama-3-8B using LoRA/QLoRA → 3–4x lower cost than GPT-4 while matching/exceeding quality
- Established MLOps best practices: GitHub Actions CI/CD, Prometheus + Grafana monitoring, feedback loops, GDPR compliance, A/B testing
- Delivered executive ROI/KPI reporting; 40%+ reduction in manual audits, 3x faster incident response
- Mentored team of 10 engineers/data scientists across India and APAC time zones

Aug 2022 – **Artificial Intelligence Engineer, Kami Vision, Bengaluru, India**

- May 2023 - Developed contextual object detection on ARM NPU (Ethos-U); outperformed YOLOv5/v8 by 18% mAP in senior-care environments
- Built automated video analysis pipeline using LLMs for captioning → 85% reduction in human review
- Managed Kubernetes inference fleet; quantization, distillation, dynamic batching → 60% monthly compute savings

- Mar 2022 – **Senior Research Associate**, *Nanyang Technological University*, Singapore
- Aug 2022 –
- Led SOTA multimodal data fusion pipeline; millions of inferences/min using PySpark and MLflow
 - Designed real-time anomaly detection system using PySpark streaming + Redis ultra-low-latency caching
 - Productionized architecture on AWS SageMaker + Kubernetes with industry partners
- Sep 2020 – **Senior Data Scientist**, *Claim Genius*, Hyderabad, India
- Mar 2022 –
- Built end-to-end pipeline extracting structured damage assessments from unstructured car images + PDF reports (OCR + LayoutParser + fine-tuned Donut/BERT)
 - Optimized CNNs (EfficientDet, ResNeXt) → 3x inference speedup on mobile GPUs
 - Integrated regression models into Django production web app
- Nov 2018 – **AI Software Developer**, *Resson (Canada-based AgriTech)*, Remote (India)
- Sep 2020 –
- Designed drone multispectral + satellite (Sentinel-2, Planet) pipeline for yield prediction, pest detection, water logging, winter-damaged canopies (22% higher accuracy)
 - Built PyTorch semantic segmentation & change detection models on TB-scale imagery using NDVI/NDWI/thermal bands
 - Developed automated orthomosaic generation and real-time pest/water-stress alerting system
 - Deployed on AWS SageMaker with automated retraining triggers
- Jun 2018 – **Computer Vision Engineer**, *Evolve Technologies*, Pune, India
- Nov 2018 –
- Built real-time X-ray baggage screening system using dual-energy imaging, CUDA-accelerated SVM, OpenGL 3D visualization (sub-100ms latency)
 - Implemented material discrimination pipeline from raw sensor data to structured threat reports

Education

- 2016 – 2018 **Master's Degree in Digital Signal Processing**, *Lovely Professional University*
Thesis: Hand Gesture Modeling for Virtual Reality Applications
- 2013 – 2017 **B.Tech – Electrical & Electronics Engineering**, *Lovely Professional University*

Certifications

AI Agents Fundamentals (LangChain) – 2024
 Parallel Programming with CUDA – NVIDIA
 Deep Learning Onramp – MathWorks
 Scrum Foundations

Publications

- 2018 Katiyar, P. et al. "Integrating Hand Gesture Modelling and Virtual Reality for Urban Planning Applications" – IEEE International Conference on Signal Processing and Communications (SPCOM)